

INTRODUCTION TO DATA MANAGEMENT

PROJECT REPORT

(Project Semester January-April 2025)

Crime against Women from 2001-14 and 2022

Submitted by

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Registration No-12304034

Programme and Section- BTECH(CSE) & KE005

Course Code -INT217

Under the Guidance of

Dr. Baljinder Kaur

Discipline of CSE/IT



CERTIFICATE

This is to certify that Thoufik umar bearing Registration no. 12304034 has completed INT217 project titled, "*Crime against Women from 2001-14 and 2022*" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 13/04/2024

DECLARATION

I, Thoufik umar, student of BTECH (CSE) under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 13/04/2025

Registration No. 12304034

ACKNOWLEDGEMENT

First and foremost, I would like to express my sincere gratitude to my project guide, Dr. Baljinder Kaur, for his valuable guidance, continuous support, and encouragement throughout the course of this project. His mentorship has helped me grow academically and personally.

I also extend my thanks to the faculty members and peers at Lovely Professional University for providing an environment that encourages learning and innovation.

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1. Introduction

Data science is transforming the way we understand and interact with complex datasets. In this project, we focus on analyzing a dataset sourced from the Open Government Data Platform (data.gov.in) India. Using Microsoft Excel, we create a dynamic dashboard to extract insights and visualize key statistics related to the

The primary objective of this project is to demonstrate the power of Excel as a tool for exploratory data analysis (EDA) and interactive data visualization.

2. Source of Dataset

The dataset used in this project is extracted from the Open Government Data (OGD) Platform India, specifically from a table titled:

“Crimes against Women-Detailed Analysis from 2001-14 and 2022”

Column1	STATE/UT	DISTRICT	Year	Rape	Kidnapping and Abduction	High focus	Dowry Deaths	Assault on women with intent to outrage her modesty	insult to modesty of Women	Cruelty by Husband or his relatives	Importation of Girls	Total Crimes
0	ANDHRA PRADESH	ADILABAD	2001	50	30	Extreme	16	149	34	175	0	404
1	ANDHRA PRADESH	ANANTAPUR	2001	23	30	Extreme	7	118	24	154	0	333
2	ANDHRA PRADESH	CHITTOOR	2001	27	34	Extreme	14	112	83	186	0	429
3	ANDHRA PRADESH	CUDDAPAH	2001	20	20	Extreme	17	126	38	57	0	258
4	ANDHRA PRADESH	EAST GODAVARI	2001	23	26	Extreme	12	109	58	247	0	452
5	ANDHRA PRADESH	GUNTAKAL RLY.	2001	0	0	Moderate	0	1	0	0	0	1
6	ANDHRA PRADESH	GUNTUR	2001	54	51	Extreme	7	139	129	378	0	704
7	ANDHRA PRADESH	HYDERABAD CITY	2001	37	39	Extreme	24	118	27	746	0	954
8	ANDHRA PRADESH	KARIMNAGAR	2001	56	49	Extreme	62	414	81	224	0	830
9	ANDHRA PRADESH	KHAMMAM	2001	47	30	Extreme	17	180	336	172	0	735
10	ANDHRA PRADESH	KRISHNA	2001	37	21	Extreme	10	208	72	265	0	576
11	ANDHRA PRADESH	KURNOOL	2001	29	47	Extreme	13	141	107	92	0	400
12	ANDHRA PRADESH	MAHABOOBNAGAR	2001	59	27	Extreme	14	176	41	69	0	327
13	ANDHRA PRADESH	MEDAK	2001	35	20	Extreme	26	100	25	192	0	363
14	ANDHRA PRADESH	NALGONDA	2001	35	19	Extreme	31	188	59	214	0	511
15	ANDHRA PRADESH	NELLORE	2001	46	80	Extreme	10	207	228	287	0	812
16	ANDHRA PRADESH	NIZAMABAD	2001	21	21	Extreme	19	55	15	228	0	338
17	ANDHRA PRADESH	PRAKASHAM	2001	19	12	Extreme	5	140	100	119	0	376
18	ANDHRA PRADESH	RANGA REDDY	2001	72	83	Extreme	37	113	55	421	7	716
19	ANDHRA PRADESH	SECUNDERABAD RLY.	2001	0	0	Moderate	1	0	1	0	0	2
20	ANDHRA PRADESH	SRIKAKULAM	2001	8	12	Extreme	6	38	47	108	0	211
21	ANDHRA PRADESH	VIJAYAWADA	2001	25	48	Extreme	2	84	122	520	0	776
22	ANDHRA PRADESH	VIJAYAWADA RLY.	2001	1	0	Moderate	0	1	1	0	0	2
23	ANDHRA PRADESH	VISAKHA RURAL	2001	12	12	Extreme	3	67	48	99	0	229
24	ANDHRA PRADESH	VISAKHAPATNAM	2001	13	6	Extreme	0	33	462	204	0	705
25	ANDHRA PRADESH	VIZIANAGARAM	2001	8	2	High	0	40	22	121	0	185
26	ANDHRA PRADESH	WARANGAL	2001	53	25	Extreme	52	241	0	165	0	483
27	ANDHRA PRADESH	WEST GODAVARI	2001	61	21	Extreme	15	246	56	348	0	686
28	ANDHRA PRADESH	TOTAL	2001	871	765	Extreme	420	3544	2271	5791	7	12798
29	ARUNACHAL PRADESH	CHANGLANG	2001	1	2	Moderate	0	3	0	0	0	5
30	ARUNACHAL PRADESH	DIBANG VALLEY	2001	2	4	Moderate	0	4	0	2	0	10
31	ARUNACHAL PRADESH	KAMENG EAST	2001	2	5	Moderate	0	5	0	0	0	10
32	ARUNACHAL PRADESH	KAMENG WEST	2001	0	2	Moderate	0	0	0	0	0	2
33	ARUNACHAL PRADESH	LOHIT	2001	2	6	High	0	8	0	0	0	14
34	ARUNACHAL PRADESH	PAPUM PARE	2001	9	5	High	0	6	0	3	0	14
35	ARUNACHAL PRADESH	SIANG EAST	2001	5	7	High	0	7	0	2	0	16
36	ARUNACHAL PRADESH	SIANG UPPER	2001	2	5	Moderate	0	3	0	0	0	8
37	ARUNACHAL PRADESH	SIANG WEST	2001	4	10	High	0	25	2	2	0	39
38	ARUNACHAL PRADESH	SUBANSIRI LOWER	2001	2	5	Moderate	0	4	0	0	0	9
39	ARUNACHAL PRADESH	SUBANSIRI UPPER	2001	2	1	Moderate	0	8	0	2	0	11
40	ARUNACHAL PRADESH	TAWANG	2001	1	0	Moderate	0	5	1	0	0	6
41	ARUNACHAL PRADESH	TIRAP	2001	1	3	Moderate	0	0	0	0	0	3

3. EDA Process

Exploratory Data Analysis (EDA) is the first step in understanding the dataset and identifying key trends, patterns, and outliers. The following steps were performed:

- Importing and cleaning data
- Universal Analysis which includes: Year-Wise Trend, Category-Wise distribution and Top states/districts of crime
- Handling missing or redundant values
- Grouping and summarizing based on key fields based on crime activities based on states
- Creating pivoted tables and charts for visual representation

Excel functions like (if and else case conditions), Sum & OR and Pivot Tables were extensively used for this purpose.

Key Objectives:

To utilize data analytics to uncover trends, patterns, and hotspots of crimes against women, with the goal of informing policies, increasing public awareness, and promoting data-backed interventions for women's safety.

1. Analyze crimes against women based on Kidnapping, Abduction and Rape using conditional logic (if, else), enabling the severity like Extreme, Moderate and High.
2. To analyze and compare the distribution of various crimes against women across the states over the year using sum.
3. Find out the top 10 States with Total crimes, which counts the severity like Extreme, High and Moderate using Line Chart.
4. To classify crimes against women based on severe atrocity (rapes), and detailed evaluation of state and year wise report using Bar graph.
5. Find out the top 10 States, Which commits the crime on Assault on Women, Cruelty Dowry Deaths, Kidnapping and Abduction. Show the information on Chart.

4. Analysis on Dataset

i. Introduction

The main analysis focused on Crime against Women (2001-14 and 24).

We aimed to determine trends such as:

- National Trend
- 2022 Spike : Likely due to improved reporting ,digital complaint platforms, and increased awareness
- Dowry Deaths, Assault and Cruelty ii. General Description The dataset includes:
 - Kidnapping , Rape and Abduction
 - Dowry Deaths , Assault on Women
 - Insult to Modesty and Cruelty by Husband or Relatives
 - Total Crime against Women iii. Specific Requirements, Functions, and Formulas

We utilized the following Excel functions and tools:

- Pivot Tables: For summarizing population by state and gender
- Slicers: To filter data interactively
- Charts: Column, bar, pie, and stacked graphs for gender ratio and birth origin
- Conditional Formatting: To highlight population thresholds iv. Analysis of Results

Key findings included:

- States like Uttar Pradesh, Rajasthan, Madhya Pradesh, and Maharashtra consistently report high figures in multiple crime categories.
- District-level analysis reveals urban and semi-urban areas with higher population density often report more cases.
- 2001-2014: Gradual but clear growth in reported crimes.

- 2022: Spike in total crimes; could indicate improved awareness, reporting mechanisms, and population growth.

v. Visualization

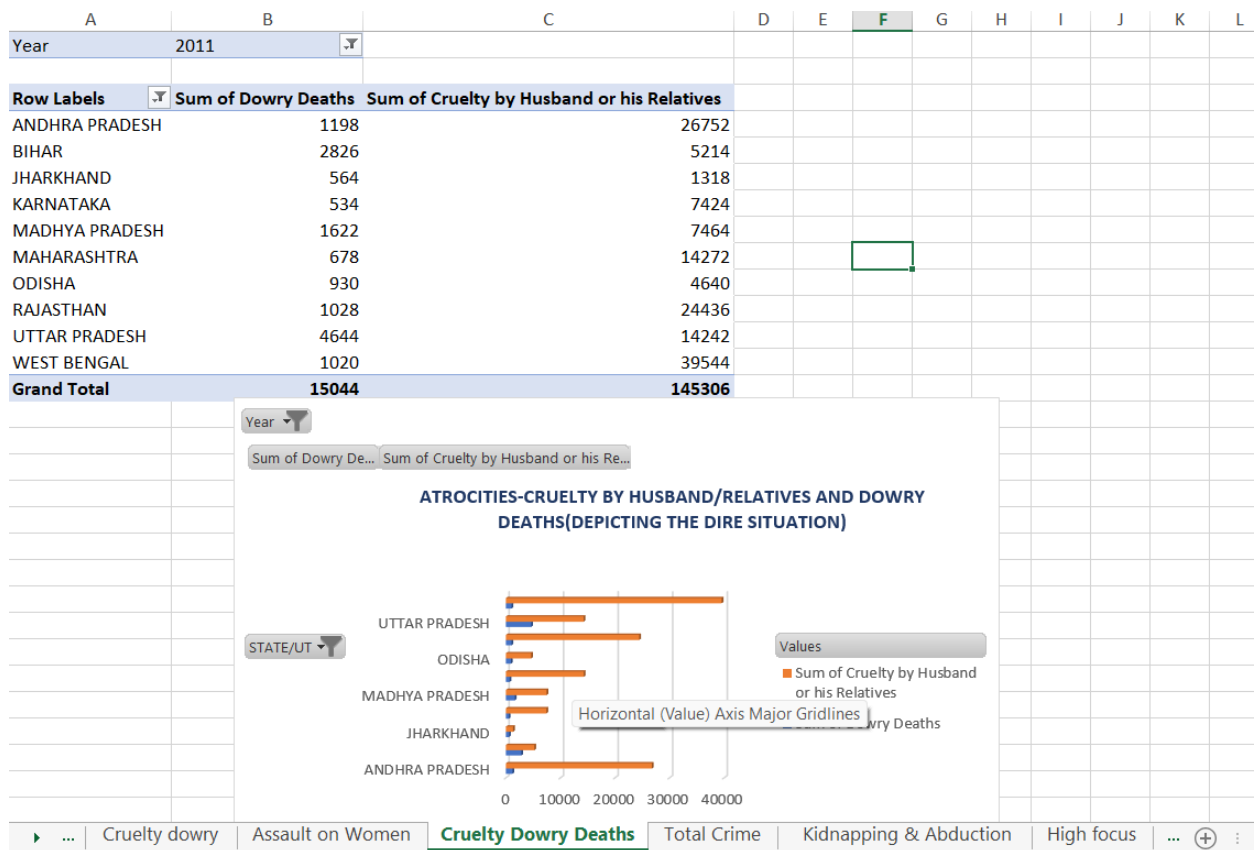
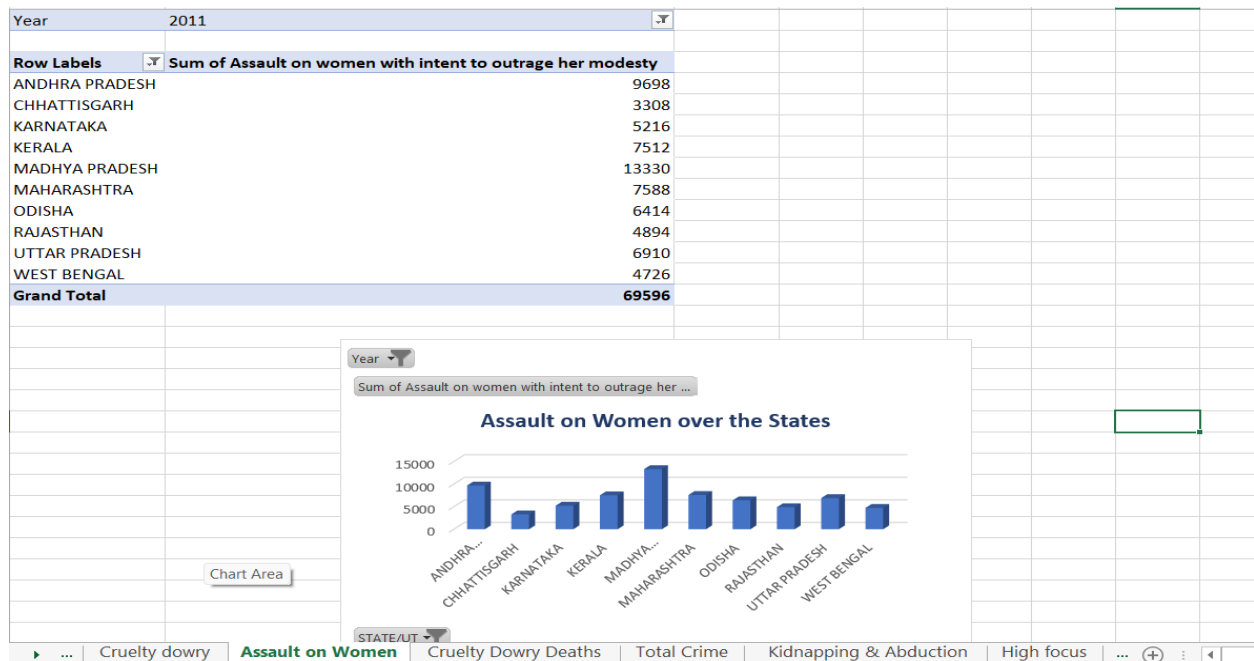
The Excel dashboard includes:

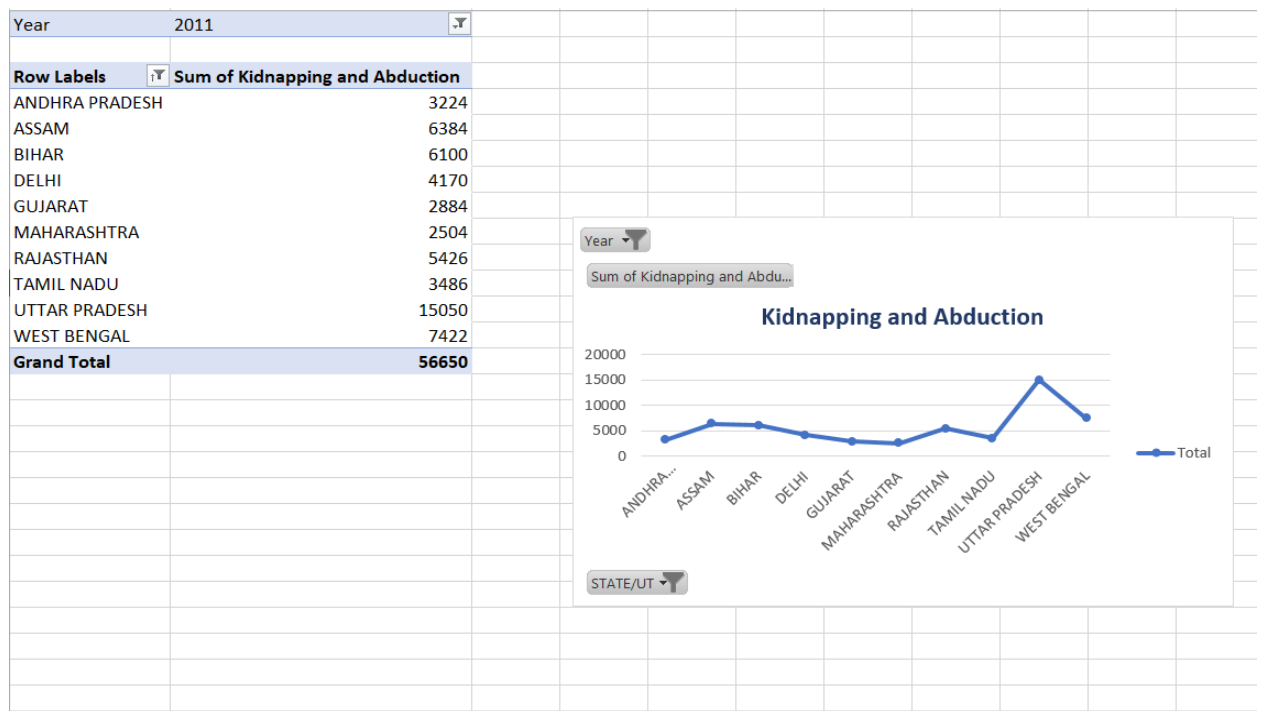
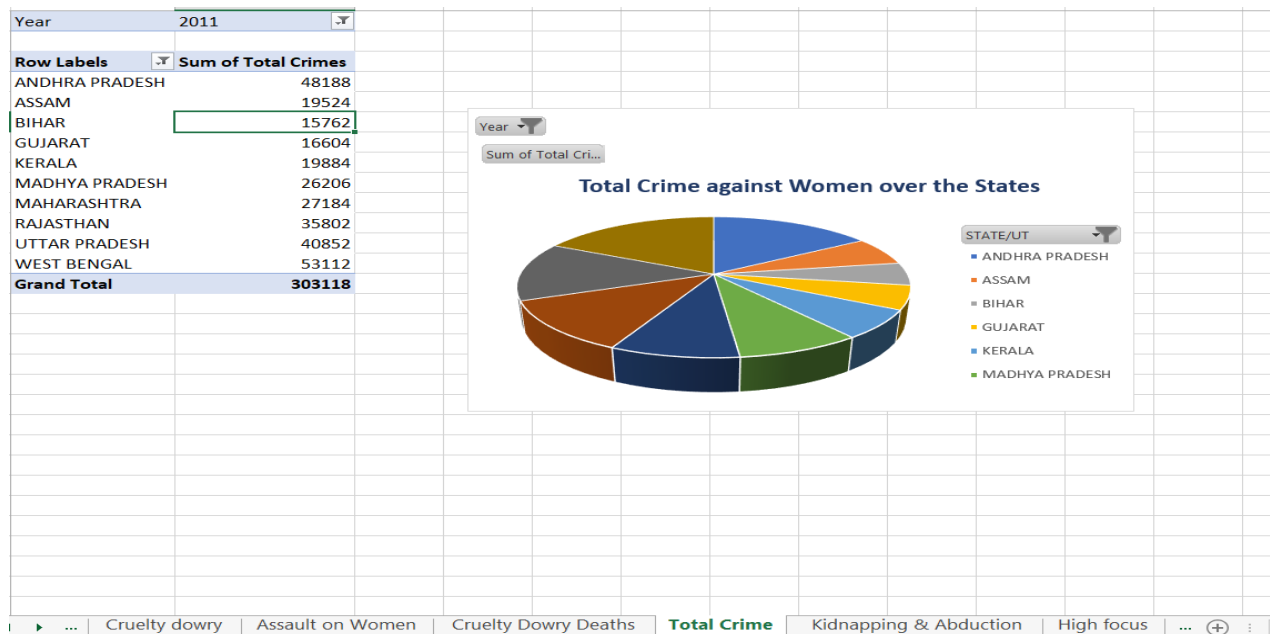
- Interactive filters and slices for different years
- Graphs showing and Comparative charts for Crime against Women

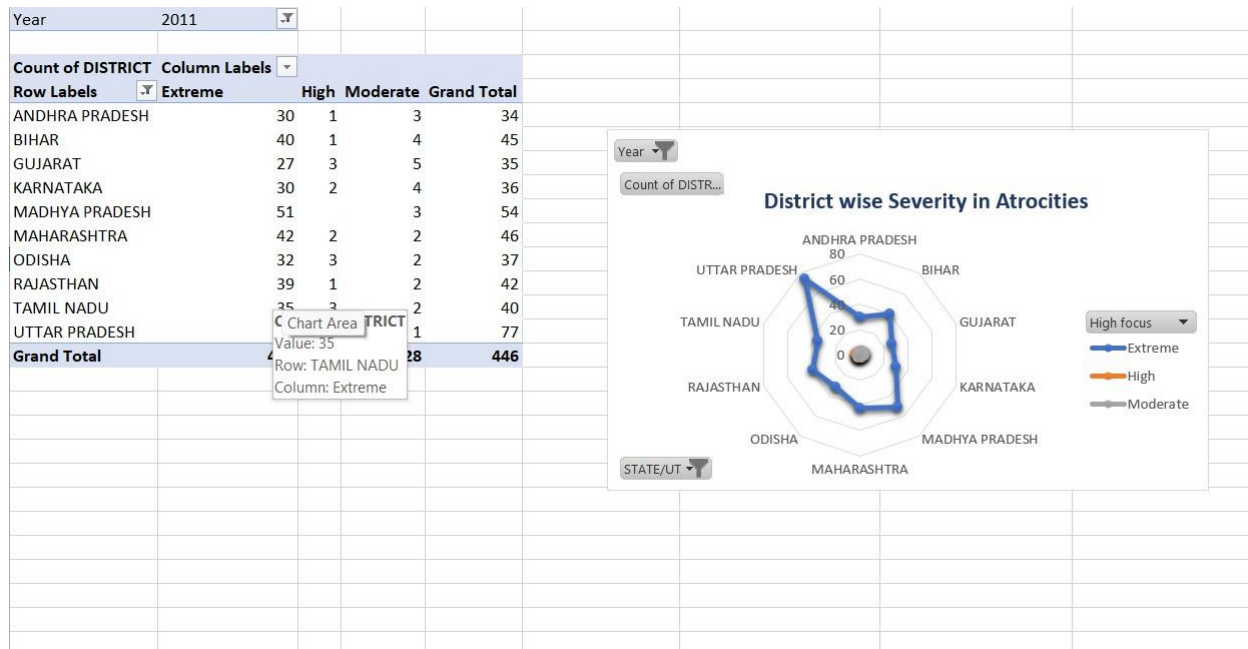
Overview of my Objectives:

Column1	STATE/UT	DISTRICT	Year	Rape	Kidnapping and Abduction	High focus	Dowry Deaths	Assault on women with intent to outrage her modesty	Insult to modesty of Women	Cruelty by Husband or his Relatives	Importation of Girls	Total Crimes
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40	ARUNACHAL PRADESH	TAWANG	2001	1	0	Moderate	0	5	1	0	0	6
41	ARUNACHAL PRADESH	TIBET	2001	1	3	Moderate	0	0	0	0	0	3

Visual Representation:







5. Conclusion

The analysis of crime data against women from 2001–2014 and in 2022 reveals a consistent and alarming rise in reported cases, highlighting both increased awareness and persistent threats to women's safety. Key crimes such as domestic violence, assault, and rape remain high, reflecting deep-rooted societal challenges. The 2022 spike emphasizes the urgent need for stronger legal

- Visualize the crime against women from 2001-14 and 2022
- Data Highlights the gap between law and implementation
- Indicates systematic societal and enforcement challenges

MY DASHBOARD



The project successfully demonstrated the usefulness of Excel for performing EDA and building informative dashboards without the need for programming. These visual insights could be valuable for law, society, and gender discrimination against women.

6. Future Scope

While this project uses Excel for analysis, there are several ways it can be extended:

- Extend data range by including upcoming years (2023 onward) for a more current trend analysis.
 - State-wise and district-level comparisons to identify high-risk zones more accurately.
 - Integration of interactive dashboards using Power BI or Tableau for better visualization.
 - Use predictive analytics or time series forecasting to project future crime trends.
 - Compare India's data with global crime statistics for broader insights.
 - Include victim demographics (age, location, relationship to offender) for deeper context.
 - Collaborate with legal and social experts to propose data-driven policy interventions.
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7. References

1. Open Government Data (OGD) Platform India: data.gov.in

Released Under: National Data Sharing and Accessibility Policy (NDSAP)

<https://www.data.gov.in/catalog/crime-against-women>

2. LinkedIn: https://www.linkedin.com/posts/thoufik_exceldashboard-dataanalytics-crimeagainstwomen-activity-7317249928685068289-Nydn?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEgE3XYBeX5VYq8Fds_x_FRTnbNXXN1nIAY

3. Kaggle: <https://www.kaggle.com/datasets/balajivaraprasad/crimes-against-women-in-india-2001-2021>